

M1S1 | Base Recap

Basic Concepts

Indentation

Python uses indentation to define blocks of code

Operators

+, -, *, /, //, %, **, @, =, +=, ^=, >, or, and, is

Data Types

int, float, bool, str, list, tuple, set, dict

Conditionals, Loops, Funcs

- if, elif, else
- for, while
- def

Mutability

each object has type, id, value and methods

```
1 x = 5
2
3 print(type(x))
4 print(id(x))
5 print(x)
6 print(dir(x))
```

Run  main ×

  | 

↑ /Users/a89088/PycharmProjects/PythonProject/.venv/b
↓ <class 'int'>
⇌ 4333707760
⇌ 5
⇌ ['__abs__', '__add__', '__and__', '__bool__', '__ce
🖨
🗑 Process finished with exit code 0

Mutable vs Immutable Objects

```
1 lst = [1, 2, 3]
2 print(id(lst))
3
4 lst.append(5)
5 print(id(lst))
```

Run main ×

↑ /Users/a89088/PycharmProjects/PythonProject/.venv/bin/python /U
4374088064
↓ 4374088064

Mutables

with editing current object you do NOT create the new one: list, dict, set

```
1 x = 5
2 print(id(x))
3
4 x += 2
5 print(id(x))
```

Run main ×

↑ /Users/a89088/PycharmProjects/PythonProject/.venv/bin/python /U
4318978544
↓ 4318978608

Process finished with exit code 0

Immutable

with editing current object, you DO create the new one: int, float, str, bool, tuple, (frozenset)

Type Annotations

functionality that allows you to explicitly specify the types of variables, parameters, and return values of functions

```
1 x: int = 5
2 name: str = "Egor"
3 def is_prime() -> bool:
4     if len(dividers) == 2:
5         return True
6      return False
```

Linters

are for code static analysis: potential bugs, PEP8 violation, general recommendations

Pylint

<https://github.com/pylint-dev/pylint>

MyPy

<https://github.com/python/mypy>

Black

<https://github.com/psf/black>